

ANSWERS OF MODEL TEST PAPER 5

INTERMEDIATE: GROUP – II

PAPER – 6: FINANCIAL MANAGEMENT & STRATEGIC MANAGEMENT

PAPER 6A : FINANCIAL MANAGEMENT

Suggested Answers/ Hints

PART I – Case Scenario based MCQs

1. (d)

2. (b)

Particulars	Computation	Result
Sales	$100 \times 5,00,000$	5,00,00,000
Less Variable cost	$100 \times 4,50,000$	4,50,00,000
Contribution		50,00,000
Less Fixed cost		25,00,000
EBIT		25,00,000
Less Interest	$15\% \times 40,00,000$	6,00,000
EBT		19,00,000

Operating leverage = $\text{Contribution} \div \text{EBIT} = 50 \text{ Lakhs} \div 25 \text{ Lakhs} = 2$ times

Operating leverage = $\% \text{ Change in EBIT} \div \% \text{ Change in Sales}$ i.e. if sales increase by 10%, EBIT increase by 20%.

Financial leverage = $\text{EBIT} \div \text{EBT} = 25 \text{ Lakhs} \div 19 \text{ Lakhs} = 1.315$ times

Combined leverage = Operating leverage $\times$ Financial leverage
 $= 2 \times 1.315 = 2.63$ times

3. (b)

Particulars	Weights	Cost in %	Weights $\times$ Cost
Share Capital	40,00,000	$5 + 1.9 \times (10 - 5) = 14.5$	5,80,000
Reserves & Surplus	25,00,000	14.5	3,62,500
Preference Share Capital	12,00,000	12	1,44,000
15% Debentures	20,00,000	$15 \times (1 - 25\%) = 11.25$	2,25,000
Total	97,00,000	Total Cost	13,11,500

Discount rate = WACC = $13,11,500 \div 97,00,000 \times 100 = 13.52\%$

4. (b)

Particulars	Computation	Result
Savings in Tea cost	200 Employees × 200 days × 3 times × ₹ 10	12,00,000
Less: Annual maintenance		(25,000)
Less: Cost of Electricity	500 units × ₹ 24 per unit × 12 months	(1,44,000)
Less: Consumables		(8,00,000)
Less: Depreciation	5,00,000 ÷ 5 years	(1,00,000)
Profit before tax		1,31,000
Less: Tax	1,31,000 × 25%	32,750
Profit after tax		98,250
Add: Depreciation		1,00,000
Cash flow after tax	98,250 + 1,00,000	1,98,250

5. (b)

Year	Particulars	Cash flow	PVF@13.52%	PV
0	Initial investment	5,00,000	1	(5,00,000)
1 to 5	Savings	1,98,250	3.473	6,88,522
	Net present value			1,88,522

6. (b) ROCE = EBIT / Total Capital Employed

Total Capital Employed = Total Assets – Current Liabilities

= 50 lakhs – 10 lakhs

= 40 lakhs

EBIT = 40 lakhs × 15%

= 6 lakhs

Now, OL of 3.5 = Contribution / EBIT

Therefore Contribution = 6 Lakhs × 3.5 = 21 lakhs

Sales = **Contribution / PV Ratio = 21 lakhs / 0.7 = 30 lakhs**

7. (d) Calculation: Cost of Debt = (Interest Payment/ Market Price of Bond)

= (8,000 / 95,000) = 8.42%

8. (d) Cost of equity will increase. As the company increases its debt ratio, the financial risk increases, which typically leads to an increase in the cost of equity as equity investors demand a higher return for the additional risk.

PART II – Descriptive Questions

1. (a) Let the EBIT at the Indifference Point level be E

Particulars	Alternative 1	Alternative 2
Description	Fully Equity of 84 Lakhs	Debt = 56 Lakhs, Equity = 28 Lakhs
EBIT	E	E
Less: Interest at 12% of ₹ 56 Lakhs	Nil	6.72
EBT	E	E – 6.72
Less: Tax at 30%	0.3 E	0.3 E – 2.016
EAT	0.7 E	0.7 E – 4.704
Less: Preference Dividend	Nil	Nil
Residual Earnings	0.7 E	0.7 E – 4.704
No. of Equity Shares (Face Value ₹ 10)	8.4 Lakh Shares	2.8 Lakh Shares
EPS = $\frac{\text{Residual Earnings}}{\text{No. of Equity Shares}}$	$\frac{0.7 E}{8.4 \text{ Lakh Shares}}$	$\frac{0.7 E - 4.704}{2.8 \text{ Lakh Shares}}$

For indifference between the above alternatives, EPS should be equal.

$$\text{So, } \frac{0.7 E}{8.4 \text{ Lakh Shares}} = \frac{0.7 E - 4.704}{2.8 \text{ Lakh Shares}}$$

On cross multiplication and simplification, $2.1 E - 14.112 = 0.7 E$. So, $1.4 E = 14.112$

$$\text{So, } E = \frac{14.112}{1.4} = 10.08$$

So, for same EPS, required EBIT = ₹ 10.08 Lakhs. EPS at that level = ₹ 0.84

Note: Presentation of solution may differ.

- (b) **Computation of PV of Future Cash Flows**

Year	Nature	Cash Flow	DF @ 12%	DCF
1	Dividends (₹ 100 × 20%)	20	0.893	17.86
2	Dividends (₹ 100 × 20%)	20	0.797	15.94
3	Dividends (₹ 100 × 20%)	20	0.712	14.24
4	Dividends (₹ 100 × 20%)	20	0.636	12.72
5	Dividends (₹ 100 × 1.2 × 20%)	24	0.567	13.61
6	Dividends (₹ 100 × 1.2 × 20%)	24	0.507	12.17
7	Dividends (₹ 100 × 1.2 × 20%)	24	0.452	10.85
7	Net Sale Proceeds (₹ 900 × 1.2 – 5%)	1,026	0.452	463.75

	Present Value of Cash Inflows			561.14
0	Less: Initial Investment (₹ 500 + 5%)	525	1	525.00
	Net Present Value			36.14

Note: At the end of Year 4, Anand will have 1.2 Share i.e. 1 Bought Share + 1/5th Bonus Share.

- (c) i. No of Eq. Shares (before buyback) = Total Earnings (before buyback)/EPS
 $= 18,00,000 / (270/18)$
 $= 1,20,000$ shares

ii. Buyback price = 270 + 10% premium = 297

iii. No of Eq. shares (after buyback) = 1,20,000 (-) 20,000 = 1,00,000 shares

iv. Total Book Value of Equity (after buyback) = 1,00,000 X 193.20 = 1,93,20,000

Now,

Total BV of Eq. (after buyback) = Total BV of Eq.(before buyback) (-) Amt of buyback

1,93,20,000 = x (-) (20,000 X 297)

Therefore x = Total BV (before buyback)

= 2,52,60,000

BV per share (before buyback) = 2,52,60,000 / 1,20,000

= 210.50 per share

2. (a) Evaluation of Factoring Proposal -

	PARTICULARS	₹	₹
(A)	Savings (Benefit) to the firm		
	Administration Cost	45,000	45,000
	Bad Debts Cost (On Recourse basis) In House – 75 lakhs X 1% Factoring – 75 lakhs X 0.5% Net Savings in bad debts cost	(75 lakhs X 0.5%)	37,500
	Cost of Carrying Debtors Cost	(WN – 1)	1,06,750
	TOTAL		1,89,250
(B)	Cost to the Firm:		

	Factor Commission [Annual credit Sales × % of Commission]	75 lakhs X 1.5%	1,12,500
	Interest Cost on Net advances	(See WN – 1)	53,100
	TOTAL		1,65,600
(C)	Net Benefits to the Firm (A – B)		23,650

Advice: Since the savings to the firm exceed the cost due to factoring, the proposal is acceptable.

WN-1 : Calculation of Savings in Interest Cost of Carrying Debtors

(I) In house Management:

Interest Cost = Credit Sales X Avg Collection Period / 360 X Interest (%) p.a

= 75,00,000 x 60/360 x 10%

= 1,25,000

(II) If Factoring services availed: If factoring services are availed, then Sukrut Limited must raise the funds blocked in receivables to the extent which is not funded by the factor (i.e amount of factor reserve (+) amount of factor commission for 30 days (+) 20% of net advances)

Calculation of Net Advances to the firm -

Debtors = 75 lakhs x 30/360 = 6,25,000

(-) Factor Reserve = 10% of above = (62,500)

(-) Factor Commission = 1.5% of Debtors = (9,375)

Net Advance = 5,53,125

Advance from Factor = 5,53,125 x 80% = 4,42,500

Int cost on Advance from Factor = 4,42,500 x 12% = 53,100

Now, the amount that is not funded by the factor (6,25,000 - 4,42,500) needs to be funded by Sukrut Limited from overdraft facility at 10%

Therefore, Int cost on Overdraft (Cost of carrying debtors) = **1,82,500 x 10% = 18,250**

Net Savings in Interest Cost of Carrying Debtors = 1,25,000 (-) 18,250 = 1,06,750

(b) Level of investment depends on the various factors listed below:

(a) Nature of Industry: Construction companies, breweries etc. requires large investment in working capital due long gestation period.

(b) Types of products: Consumer durable has large inventory as compared to perishable products.

- (c) **Manufacturing Vs Trading Vs Service:** A manufacturing entity has to maintain three levels of inventory i.e. raw material, work-in-process and finished goods whereas a trading and a service entity has to maintain inventory only in the form of trading stock and consumables respectively.
- (d) **Volume of sales:** Where the sales are high, there is a possibility of high receivables as well.
- (e) **Credit policy:** An entity whose credit policy is liberal has not only high level of receivables but may require more capital to fund raw material purchases as that will depend on credit period allowed by suppliers.

3. (a) **WN-1 : Calculation of Cost of Debt (Kd)**

$$\text{Approximation Method} = \frac{\text{Int (1-t)} + (\text{RV} - \text{NP}) / \text{N}}{(\text{RV} + \text{NP}) / 2}$$

$$\text{RV} = 100 + 10\% = 110, \text{NP} = 105 - 4\% = 100.8$$

$$= \frac{10 (1 - 0.25) + (110 - 100.8) / 10}{(110 + 100.8) / 2} = 7.99\%$$

YTM Method:

$$\text{CMP (Po) (-) Floatation Cost} = \{\text{Int}(1-t) \times \text{PVAF (r\%, 10 years)}\} + \{\text{RV} \times \text{PVIF (r\%, 10}^{\text{th}} \text{ Year)}\}$$

$$105 - 4\% = \{10 (1 - 0.25) \times \text{PVAF (r\%, 10 years)}\} + \{110 \times \text{PVIF (r\%, 10}^{\text{th}} \text{ year)}\}$$

Using trial and error method, NPV at 5% & 10%

Year	Cash flows	Disc Factor @ 5%	PV (₹)	Disc Factor @ 10%	PV (₹)
0	-100.8	1	-100.8	1	-100.8
1 to 10	7.5	7.7217	57.91275	6.1446	46.0845
10	110	0.6139	67.529	0.3855	42.405
			24.64175		-12.3105

$$\text{IRR} = 5 + \frac{24.64175}{24.64175 - (-12.3105)} \times (10 - 5) = 8.33\%$$

$$\text{Therefore overall cost of debt (Kd)} = (7.99 + 8.33) / 2 = 8.16\%$$

WN-2 : Calculation of Cost of Preference (Kp)

$$\text{Approximation Method} = \frac{\text{Pref. Div.} + (\text{RV} - \text{NP}) / \text{N}}{(\text{RV} + \text{NP}) / 2}$$

$$\text{RV} = 100 \text{ NP} = 115 - 2\% = 112.7$$

$$= \frac{12 + (100 - 112.7) / 10}{(100 + 112.7) / 2} = 10.09\%$$

YTM Method:

CMP (Po) (-) Floatation Cost = {Pref Div × PVAF (r%,10years)} + {RV × PVIF (r%,10th Year)}

$$115 - 2\% = \{12 \times \text{PVAF (r\%,10 years)}\} + \{100 \times \text{PVIF (r\%,10}^{\text{th}} \text{ year)}\}$$

Using trial and error method, NPV at 5% & 10%

Year	Cash flows	Disc Factor @ 5%	PV (₹)	Disc Factor @ 10%	PV (₹)
0	-112.7	1	-112.7	1	-112.7
1 to 10	12	7.7217	92.6604	6.1446	73.7352
10	100	0.6139	61.39	0.3855	38.55
			41.3504		-0.4148

$$\text{IRR} = 5 + \frac{41.3504}{41.3504 - (-0.4148)} \times (10 - 5) = \mathbf{9.95\%}$$

Therefore, overall cost of debt (Kp) = (10.09 + 9.95) / 2 = 10.02%

WN-3 : Calculation of Cost of equity (Ke)

$$\text{Ke} = \{D1 / (\text{Po} - \text{Floatation})\} + G$$

$$= \{2 + 9\% / 27 - 4.5\} + 0.09$$

$$= 18.69\%$$

Calculation of WACC using market value weights

Source of Capital	Working	Market Value	Weights	Cost (K)	WACC (Ko)
		(₹)	(A)	(B)	(A × B)
Equity	27 x 150000	40,50,000	0.7377	18.69	13.7877
Reserves	Included in equity	-	-	-	-
Preference	115 x 7500	8,62,500	0.1571	10.02	1.5741
Debentures	105 x 5500	5,77,500	0.1052	8.16	0.8584
		54,90,000	1		16.22%

$$\text{WACC (Ko)} = \mathbf{16.22\%}$$

(b) **Change in Reserve & Surplus = ₹ 25, 00,000 – ₹ 20,00,000 = ₹ 5,00,000**

So, Net profit = ₹ 5, 00,000

(i) Net Profit Ratio = 8%

$$\therefore \text{Sales} = \frac{5,00,000}{8\%} = ₹ 62,50,000$$

(ii) Cost of Goods sold

$$\begin{aligned}
&= \text{Sales} - \text{Gross profit Margin} \\
&= ₹ 62, 50,000 - 20\% \text{ of } ₹ 62, 50,000 \\
&= ₹ 50, 00,000
\end{aligned}$$

$$(iii) \text{ Fixed Assets} = \frac{₹ 30,00,000}{40\%} = ₹ 75,00,000$$

$$(iv) \text{ Stock} = \frac{\text{Cost of Goods Sold}}{\text{STR}} = \frac{50,00,000}{4} = ₹ 12,50,000$$

$$(v) \text{ Debtors} = \frac{62,50,000}{360} \times 90 = ₹ 15,62,500$$

$$(vi) \text{ Cash Equivalent} = \frac{50,00,000}{12} \times 1.5 = ₹ 6,25,000$$

Balance Sheet as on 31st March 2024

Liabilities	(₹)	Assets	(₹)
Share Capital	50,00,000	Fixed Assets	75,00,000
Reserve and Surplus	25,00,000	Sundry Debtors	15,62,500
Long-term loan	30,00,000	Closing Stock	12,50,000
Sundry Creditors (Balancing Figure)	4,37,500	Cash in hand	6,25,000
	1,09,37,500		1,09,37,500

4. (a) Though in a sole proprietorship firm, partnership etc., owners participate in management but in corporates, owners are not active in management so, there is a separation between owner/ shareholders and managers. In theory managers should act in the best interest of shareholders however in reality, managers may try to maximise their individual goal like salary, perks etc., so there is a **principal agent relationship between managers and owners, which is known as Agency Problem**. In a nutshell, Agency Problem is the chances that managers may place personal goals ahead of the goal of owners. Agency Problem leads to Agency Cost. Agency cost is the additional cost borne by the shareholders to monitor the manager and control their behaviour so as to maximise shareholders wealth. Generally, Agency Costs are of four types (i) monitoring (ii) bonding (iii) opportunity (iv) structuring.

Addressing the agency problem

The agency problem arises if manager's interests are not aligned to the interests of the debt lender and equity investors. The agency problem of debt lender would be addressed by imposing negative covenants i.e. the managers cannot borrow beyond a point. This is one of the most important concepts of modern day finance and the application of this would be applied in the Credit Risk Management of Bank, Fund Raising, Valuing distressed companies.

Agency problem between the managers and shareholders can be addressed if the interests of the managers are aligned to the interests of the shareholders. It is easier said than done.

However, following efforts have been made to address these issues:

- ◆ Managerial compensation is linked to profit of the company to some extent and also with the long term objectives of the company.
- ◆ Employee is also designed to address the issue with the underlying assumption that maximisation of the stock price is the objective of the investors.
- ◆ Effecting monitoring can be done.

- (b) (i) **Sales and Lease Back:** Under this type of lease, the owner of an asset sells the asset to a party (the buyer), who in turn leases back the same asset to the owner in consideration of a lease rentals. Under this arrangement, the asset is not physically exchanged but it all happen in records only. The main advantage of this method is that the lessee can satisfy himself completely regarding the quality of an asset and after possession of the asset convert the sale into a lease agreement.

Under this transaction, the seller assumes the role of lessee (as the same asset which he has sold came back to him in the form of lease) and the buyer assumes the role of a lessor (as asset purchased by him was leased back to the seller). So, the seller gets the agreed selling price and the buyer gets the lease rentals.

- (ii) **Leveraged Lease:** Under this lease, a third party is involved besides lessor and the lessee. The lessor borrows a part of the purchase cost (say 80%) of the asset from the third party i.e., lender and asset so purchased is held as security against the loan. The lender is paid off from the lease rentals directly by the lessee and the surplus after meeting the claims of the lender goes to the lessor. The lessor is entitled to claim depreciation allowance.
- (iii) **Sales-aid Lease:** Under this lease contract, the lessor enters into a tie up with a manufacturer for marketing the latter's product through his own leasing operations, it is called a sales-aid lease. In consideration of the aid in sales, the manufacturer may grant either credit or a commission to the lessor. Thus, the lessor earns from both sources i.e. From lessee as well as the manufacturer.
- (iv) **Close-ended and Open-ended Leases:** In the close-ended lease, the assets get transferred to the lessor at the end of lease, the risk of obsolescence, residual value etc., remain with the lessor being the legal owner of the asset. In the open-ended lease, the lessee has the option of purchasing the asset at the end of the lease period.

- (c) The basic objective of financial management is to design an appropriate capital structure which can provide the highest wealth, i.e., highest MPS, which in turn depends on EPS.

Given a level of EBIT, EPS will be different under different financing mix depending upon the extent of debt financing. The effect of leverage on the EPS emerges because of the existence of fixed financial charge i.e., interest on debt, financial fixed dividend on preference share capital. The effect of fixed financial charge on the EPS depends upon the relationship between the rate of return on assets and the rate of fixed charge. If the rate of return on assets is higher than the cost of financing, then the increasing use of fixed charge financing (i.e., debt and preference share capital) will result in increase in the EPS. This situation is also known as favourable financial leverage or Trading on Equity. On the other hand, if the rate of return on assets is less than the cost of financing, then the effect may be negative and, therefore, the increasing use of debt and preference share capital may reduce the EPS of the firm.

The fixed financial charge financing may further be analysed with reference to the choice between the debt financing and the issue of preference shares. Theoretically, the choice is tilted in favour of debt financing for two reasons: (i) the explicit cost of debt financing i.e., the rate of interest payable on debt instruments or loans is generally lower than the rate of fixed dividend payable on preference shares, and (ii) interest on debt financing is tax-deductible and therefore the real cost (after-tax) is lower than the cost of preference share capital.

OR

- (c) When the cost of 'fixed cost fund' is less than the return on investment, financial leverage will help to increase return on equity and EPS. The firm will also benefit from the saving of tax on interest on debts etc. However, when cost of debt will be more than the return it will affect return of equity and EPS unfavourably and as a result firm can be under financial distress. Therefore, financial leverage is also known as **"double edged sword"**.

Effect on EPS and ROE:

When, $ROI > \text{Interest}$ – Favourable – Advantage

When, $ROI < \text{Interest}$ – Unfavourable – Disadvantage

When, $ROI = \text{Interest}$ – Neutral – Neither advantage nor disadvantage